

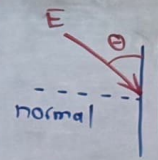
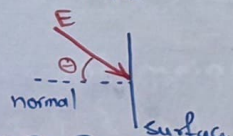
Electric Flux

Electric Flux

* The number of E lines penetrating surface.

$$\phi = \vec{E} \cdot \vec{A} \Rightarrow \phi = EA \cos \theta \text{ or } \phi = EA \sin \theta$$

Electric Flux



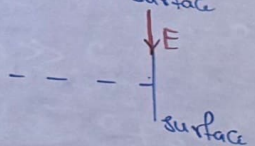
* electric flux can be three Positive or Negative total by algebraic sum

Superposition
total → Vector Sum

Maximum $\phi = \phi_{\max} = EA$

E parallel to normal or E is perpendicular to surface

Minimum $\phi = \phi_{\min} = 0$



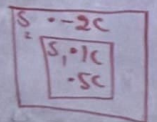
E perpendicular to normal or E is parallel to surface

$\phi = \phi_{\max} \cos \theta$

Gauss's law

* electric flux through any closed surface is:
electric flux depends on the charge inside surface
electric flux is independent of a shape of surface

$$\phi = \frac{Q_{\text{in}}}{\epsilon_0}$$

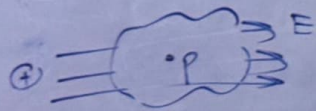


$$\phi_{\text{in}} = \frac{1+5}{\epsilon_0} = 6/\epsilon_0$$

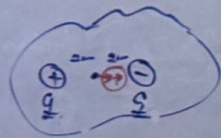
$$\phi_{\text{out}} = \frac{5+1-2}{\epsilon_0} = 4/\epsilon_0$$

① Out E lines are more than in E lines = positive flux
" " " equal " " " " : zero flux
" " " less " " " " : Negative flux

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$$E = k_e \frac{Q}{r^2}$$



0C



2C

